Power series basics

We likely would have learned about power series in a Calculus 2 course, but since that may have been a while ago, let's do a quick review of them here.

Since the series solution method we're about to learn is entirely based around power series, it's worth making sure that we're starting with a good foundation.

Power series

A **power series centered at** x_0 is given by

$$\sum_{n=0}^{\infty} c_n(x-x_0)^n = c_0 + c_1(x-x_0) + c_2(x-x_0)^2 + c_3(x-x_0)^3 + \dots$$

We also call this a power series in $x - x_0$. We'll particularly focus on power series centered at $x_0 = 0$ (or a power series in x), in which case our power series expression simplifies to

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

In this power series, $n = 0$ is the **index**. The index gives the starting value of n, while the ∞ we see above the summation notation gives us the "ending" value of n. In other words, if we read the summation from bottom to top, from $n = 0$, to the summation, to the ∞, it tells us to add all terms, starting at $n = 0$, until $n = \infty$.

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The index can start at any value, but we'll most commonly see indices for $n = 0$, $n = 1$, or $n = 2$.

Convergence of power series

A power series is **convergent** at a specific value of x when the limit exists,

$$\lim_{N\to\infty} \sum_{n=0}^{N} c_n(x - x_0)^n$$

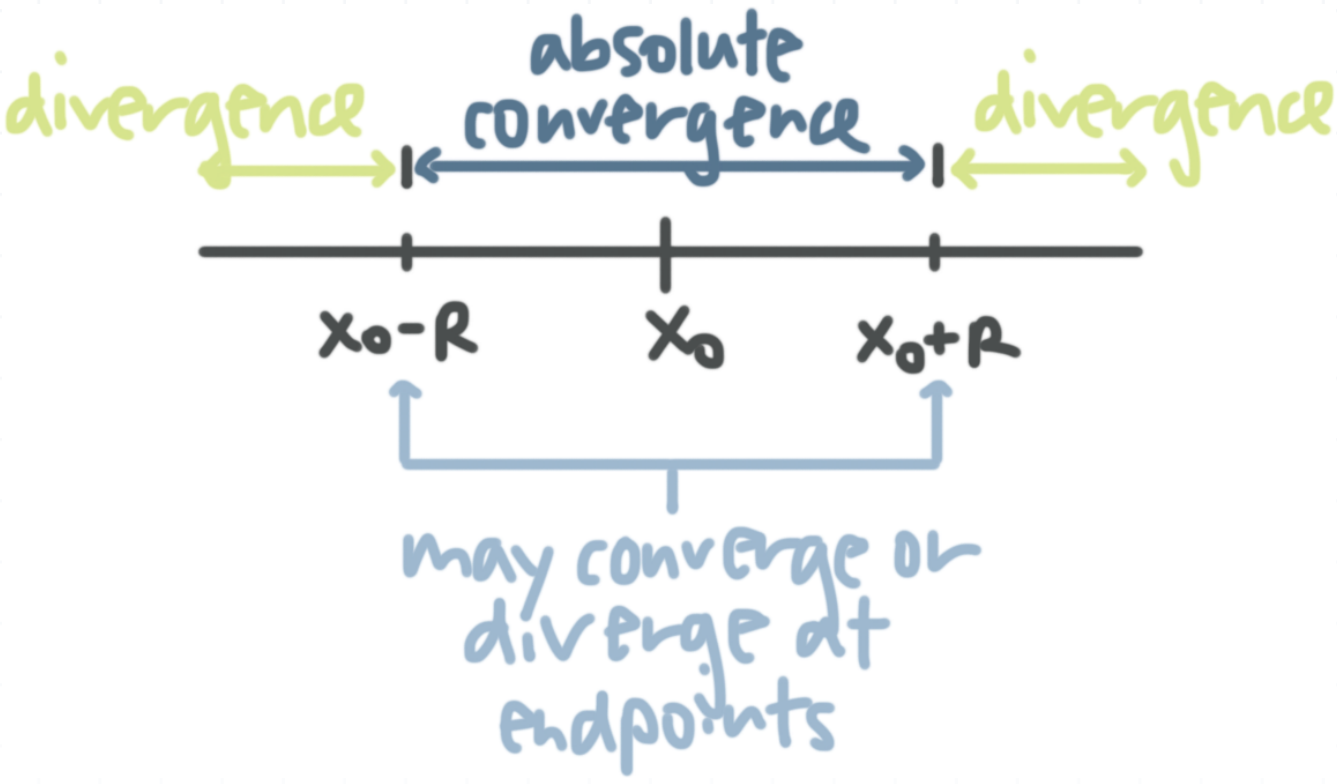
Otherwise, if this limit does not exist, then the series is divergent. For a power series centered at x_0, the **interval of convergence** of the series is the set of all real numbers x around x_0 for which the series converges, and the radius R of the interval of convergence is the **radius of convergence**.

If $R > 0$, then the power series converges for $|x - x_0| < R$ and diverges for $|x - x_0| > R$. If $R = 0$, then the series converges only at x_0, its center. When $R = \infty$, it means the power series converges for all x.

The power series converges absolutely inside the interval of convergence, diverges outside the interval, and may or may not converge at the endpoints of the interval.

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We can often use the ratio test to determine the convergence of a power series. After calculating this value L,

$$L = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right|$$

the **ratio test** tells us that the series converges absolutely if $L < 1$, diverges if $L > 1$, and inconclusive if $L = 1$. The ratio test will always be inconclusive at the endpoints of the interval of convergence, $x_0 \pm R$.

Let's do an example so that we understand how to find the interval and radius of convergence.

Example

Find the interval and radius of convergence of the power series.

$$\sum_{n=1}^{\infty} \frac{2^{5n}}{5^{2n}} \left(\frac{x}{3} \right)^n$$

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First, let's rewrite the series.

$$\sum_{n=1}^{\infty}\left(\frac{2^5}{5^2}\right)^n\left(\frac{x}{3}\right)^n$$

$$\sum_{n=1}^{\infty}\left(\frac{32}{25}\right)^n\left(\frac{x}{3}\right)^n$$

Then applying the ratio test to the power series gives

$$L=\lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|$$

$$L=\lim_{n\to\infty}\left|\frac{\left(\frac{32}{25}\right)^{n+1}\left(\frac{x}{3}\right)^{n+1}}{\left(\frac{32}{25}\right)^{n}\left(\frac{x}{3}\right)^{n}}\right|$$

$$L=\lim_{n\to\infty}\left|\left(\frac{32}{25}\right)\left(\frac{x}{3}\right)\right|$$

$$L=\lim_{n\to\infty}\left|\frac{32}{75}x\right|$$

$$L=\frac{32}{75}|x|$$

To determine where the series converges absolutely, we set this value $L<1$.

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$$\frac{32}{75}|x| < 1$$

$$|x| < \frac{75}{32}$$

$$-\frac{75}{32} < x < \frac{75}{32}$$

So the interval of convergence is $(-75/32, 75/32)$, and the radius of convergence is $R = 75/32$. To determine whether the series converges or diverges at the endpoints, we would need to substitute them into the original series.

For $x = -75/32$,

$$\sum_{n=1}^{\infty} \frac{2^{5n}}{5^{2n}} \left(\frac{-\frac{75}{32}}{3} \right)^n = \sum_{n=1}^{\infty} \frac{2^{5n}}{5^{2n}} \left(\frac{-\frac{3^n \cdot 5^{2n}}{2^{5n}}}{3^n} \right) = \sum_{n=1}^{\infty} (-1)$$

For $x = 75/32$,

$$\sum_{n=1}^{\infty} \frac{2^{5n}}{5^{2n}} \left(\frac{\frac{75}{32}}{3} \right)^n = \sum_{n=1}^{\infty} \frac{2^{5n}}{5^{2n}} \left(\frac{\frac{3^n \cdot 5^{2n}}{2^{5n}}}{3^n} \right) = \sum_{n=1}^{\infty} 1$$

Then we can see that the series is divergent at both endpoints, and therefore that the interval of convergence is still $(-75/32, 75/32)$. If the series had converged at both endpoints, the interval would have been $[-75/32, 75/32]$. If the series were convergent at just the left endpoint, or convergent at just the right endpoint, then the interval of convergence would have been $[-75/32, 75/32)$ or $(-75/32, 75/32]$, respectively.

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Functions as power series

We can define a function as a power series

$$f(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^n$$

The domain of the function is the interval of convergence of the series. If the radius of convergence is $R > 0$ or $R = \infty$, then the function f is continuous, differentiable, and integrable on $(x_0 - R, x_0 + R)$ or $(-\infty, \infty)$, respectively.

We say that a function is **analytic at a point** x_0 if the function can be represented by a power series in $x - x_0$ with any positive or infinite radius of convergence.

A **Taylor series** is just a power series in which the coefficients c_n are given by $(f^{(n)}(x_0))/n!$, where $f^{(n)}$ are the derivatives of f. So the Taylor series representation of a function f is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

$$= \frac{f^{(0)}(x_0)}{0!}(x - x_0)^0 + \frac{f^{(1)}(x_0)}{1!}(x - x_0)^1 + \frac{f^{(2)}(x_0)}{2!}(x - x_0)^2 + \frac{f^{(3)}(x_0)}{3!}(x - x_0)^3 + \dots$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

$$= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \dots$$

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The **Maclaurin series** is simply the Taylor series centered at $x_0 = 0$, so the Maclaurin series representation of f is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

The common Maclaurin series in the table below will come in handy as we talk about series solutions to differential equations.

Maclaurin series	Interval of convergence
$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$	$(-\infty, \infty)$
$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$	$(-\infty, \infty)$
$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$	$(-\infty, \infty)$
$\tan^{-1} x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$	$[-1,1]$
$\cosh x = \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n}$	$(-\infty, \infty)$
$\sinh x = \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1}$	$(-\infty, \infty)$
$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n$	$(-1,1]$

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$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \qquad (-1,1)$$

Let's do a quick example so that we can see how to adapt these common Maclaurin series to a similar function.

Example

Find the Maclaurin series representation of $f(x) = \sin(2x^2)$.

Because we already have the Maclaurin series representation of $f(x) = \sin x$, we can simply replace x with $2x^2$.

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$\sin(2x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (2x^2)^{2n+1}$$

$$\sin(2x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (2^{2n+1} x^{2(2n+1)})$$

$$\sin(2x^2) = \sum_{n=0}^{\infty} \frac{2^{2n+1}(-1)^n}{(2n+1)!} x^{4n+2}$$

This is the Maclaurin series representation of $f(x) = \sin(2x^2)$, and the interval of convergence is still $(-\infty, \infty)$, the same as the interval of convergence for $f(x) = \sin x$.

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Differentiation of power series

Finally, we mentioned before that if the radius of convergence is positive, then the function is differentiable on the entire interval of convergence.

With that in mind, it'll be important to be able to find the first and second derivatives of a power series in x, which are

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$$

$$y'' = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + \dots$$

We'll use these very often in upcoming lessons as we start working on finding series solutions to differential equations.

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